

ECE 647: Lecture 6

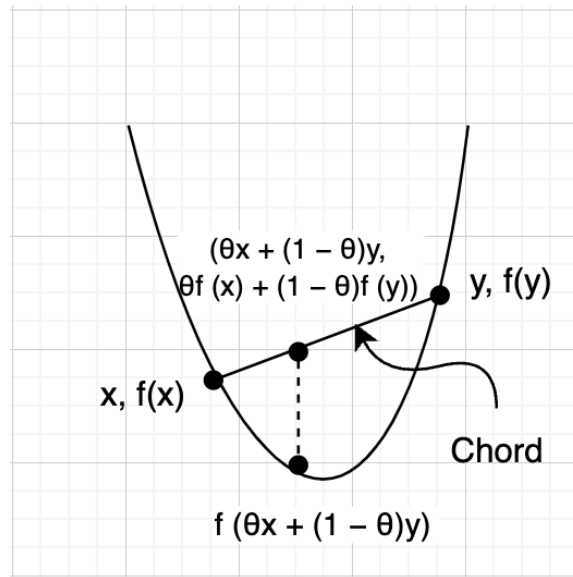
Convex Functions

A function $f : \mathcal{R}^n \rightarrow \mathcal{R}$ is convex if its domain $\text{dom}(f)$ is a convex set and

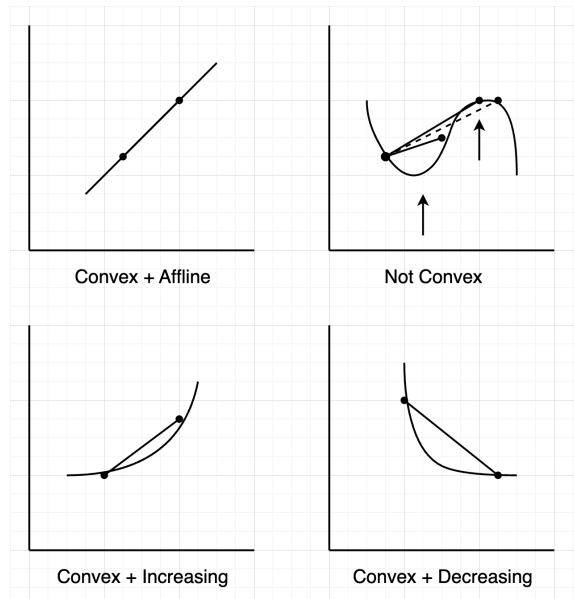
$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y) \quad (1)$$

for all $x, y \in \text{dom}(f)$, $0 \leq \theta \leq 1$

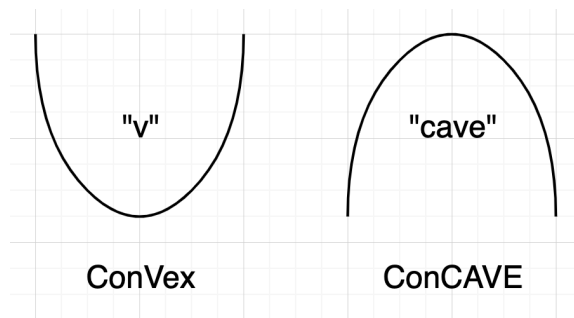
Geometrical Interpretation - The line segment between $(x, f(x))$ and $(y, f(y))$, which is the chord between x and y , lies above the graph of f .



Affine functions are convex by definition. Equation (1) is satisfied with equality.



f is concave if $-f$ is convex.



Affine functions are both convex and concave.

A function is strictly convex if strict inequality holds in equation 1 whenever $x \neq y$ and $0 < \theta < 1$ – Does not contain any affine parts.

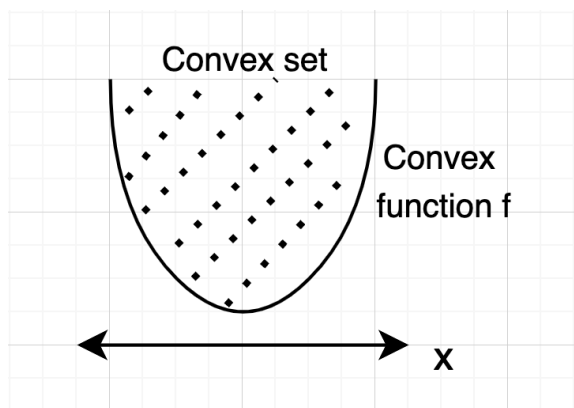
Relationship between Convex Functions and Convex Sets

0.1 Epigraph

Define the epigraph of a function $f : R^n \rightarrow R$ as

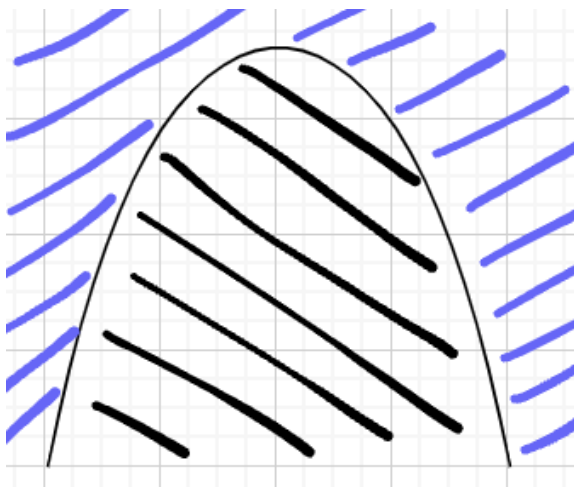
$$epi(f) = \{(x, t) | x \in dom(f), t \geq f(x)\} \quad (2)$$

A function is convex iff its epigraph is a convex set.



Proof by definition. This will be useful in proving strong duality.

- Q - Is the epigraph of a concave function a convex set?

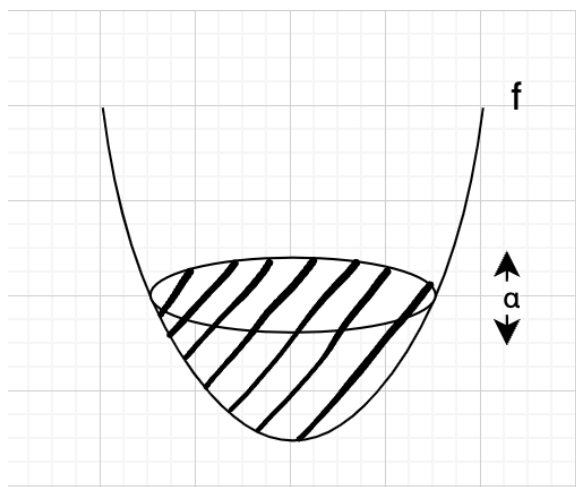


$$epi(f) = \{(x, t) | x \in dom(f), t \leq f(x)\} \quad (3)$$

0.2 Sublevel Set

Define the sublevel set

$$C(\alpha) = \{x \in dom(f) | f(x) \leq \alpha\} \quad (4)$$

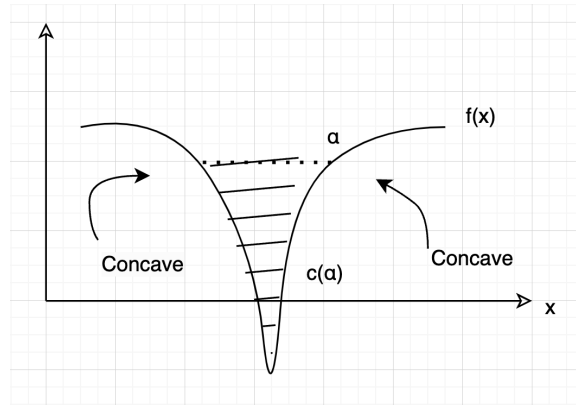


If f is a convex function, then $C(\alpha)$ is a convex set for any α . Proof by definition. This will be useful in proving strong duality.

- Q - Is the sublevel set of a concave function a convex set?

$$\{x | f(x) \geq \alpha\} \quad (5)$$

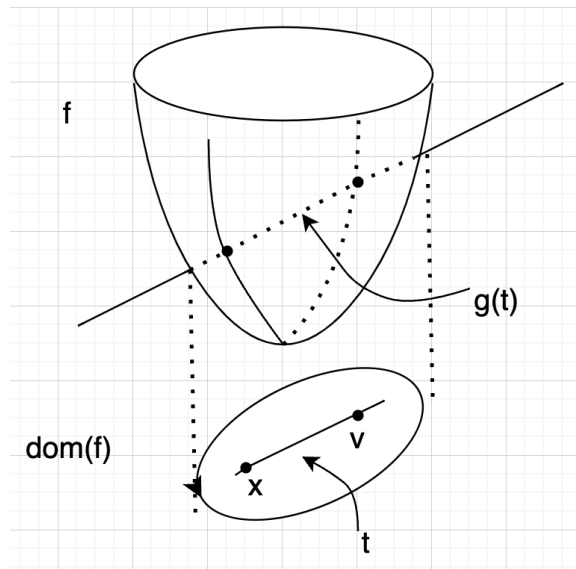
- Q - Is the converse true? In other words if the sublevel set $C(\alpha)$ of a function f is convex for any α , then is the function f convex? A - No



Restriction to a Line

A function is convex if and only if it is convex when restricted to any line that intersects its domain.

f is convex $\iff$ for all $x \in \text{dom}(f)$ and all v , $g(t) = f(x + tv)$ is convex on $\{t | x + tv \in \text{dom}(f)\}$



- This is not surprising because, to check convexity, we are only checking on straight lines anyways.

- A function is convex if, for any two points, the chord is above the function. But the chord is also above that part of the function restricted to the same line.
- This is useful because we can reduce the problem of checking the convexity of any function to that of checking the convexity of functions of one variable.
- Q - A function $f(x,y)$ is convex wrt each variable (x or y), then is f a convex function of x and y ? A - No. Example - $f(x,y) = xy$. Product of convex functions may not be convex.

Conditions for Convexity

Often it is not easy to verify convexity of a function f using the definition directly.

The following conditions will be handy -

- First Order Conditions ∇f
- Second Order Conditions $\nabla^2 f$
- Conditions that preserve convexity

First Order Conditions

- If f is differentiable in its domain $\text{dom}(f)$, then f is convex iff $\text{dom}(f)$ is a convex set and

$$f(y) \geq f(x) + [\nabla f(x)]^T (y - x) \quad (6)$$

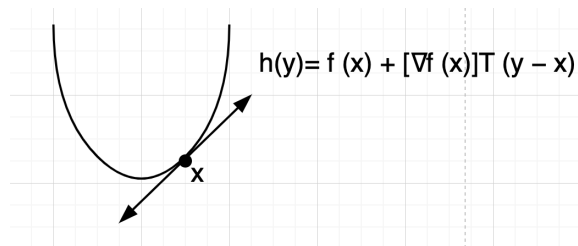
for all $x, y \in \text{dom}(f)$

- f is strictly convex iff $\text{dom}(f)$ is a convex set and

$$f(y) > f(x) + [\nabla f(x)]^T (y - x) \quad (7)$$

for all $x \neq y, x, y \in \text{dom}(f)$

- Note that, $f(x) + [\nabla f(x)]^T (y - x)$ defines a hyperplane(of the variable $y - p(y)$) tangent to $f(x)$ at $(x, f(x))$



- Hence the condition states that the rest of the function must be above the hyperplane - "supporting hyperplane"
- A similar result holds for concave functions

